COURSE SYLLABUS

APPLIED STATISTICS (307312)

Master's level | Student Form | Draft

Course information	Details
Course no. / title	307312 / Applied Statistics
Academic level	Master's (graduate)
Academic year / semester	[To be confirmed]
Credit hours / delivery mode	[Confirm from approved course specification]
Prerequisite / co-requisite	[To be confirmed]
Instructor	Dr. Mohammed Fasha [confirm assignment]
Email / office	mohammed.fasha@uop.edu.jo / [confirm office]
LMS	Moodle: https://lms.uop.edu.jo
Office hours	[Announce on Moodle]

Course Description

Beginning with an introduction to R and RStudio, this graduate-level course develops a practical foundation in applied statistics for evidence-based decision-making. It covers data collection and visualization; descriptive statistics; probability and probability distributions; sampling, estimation and hypothesis testing; comparison of groups; analysis of variance; chi-square and nonparametric methods; simple and multiple linear regression; time-series forecasting; and business analytics. Students use R to analyze real datasets, evaluate assumptions, interpret uncertainty, and communicate reproducible findings.

Course Objectives

• Develop statistically sound approaches to collecting, preparing, describing and visualizing data.

• Apply probability and sampling theory to quantify variability and uncertainty.

• Select, conduct and interpret estimation, hypothesis testing and comparison procedures.

• Build, assess and communicate regression and introductory forecasting models.

• Use R to develop reproducible analyses and actionable evidence for business decisions.

Course Intended Learning Outcomes (ILOs)

ILO	Outcome	Teaching and assessment
K1	Explain variable types, sampling, descriptive summaries, probability models and sampling distributions.	Interactive lecture; R exercises; quizzes/examinations; applied report
K2	Evaluate the logic, assumptions and interpretation of estimation, hypothesis testing and statistical models.	Interactive lecture; R exercises; quizzes/examinations; applied report
I1	Select appropriate statistical procedures for a stated business or research question and justify choices.	Interactive lecture; R exercises; quizzes/examinations; applied report
I2	Critically evaluate statistical evidence,	Interactive lecture; R exercises;

	uncertainty, model assumptions and limitations.	quizzes/examinations; applied report
P1	Prepare, visualize and summarize real-world datasets using R.	Interactive lecture; R exercises; quizzes/examinations; applied report
P2	Implement inferential procedures, regression analyses and basic forecasts using R.	Interactive lecture; R exercises; quizzes/examinations; applied report
C1	Communicate statistically defensible findings in a reproducible report for a nontechnical audience.	Interactive lecture; R exercises; quizzes/examinations; applied report

Program-ILO codes and formal alignment must be completed against the approved Master's program specification; undergraduate PILO codes from 307102 have not been copied.

Course Schedule (15 Teaching Weeks)

Week	Topic(s)	Topic details	Teaching / student tasks
1	Introduction to R and RStudio	R/RStudio setup; scripts and projects; objects, vectors, data frames; importing CSV; packages; basic functions and visualizations; reproducible workflows.	Interactive lecture / practical R lab; Install or access R and RStudio; run a script; import, inspect, summarize and plot a small dataset.
2	1. Defining and Collecting Data; 2. Organizing and Visualizing Variables	Research questions, data sources, populations and samples, measurement scales, cleaning, frequency tables and charts.	Interactive lecture / practical R lab; Inspect a dataset and create appropriate tables and visualizations in R.
3	3. Numerical Descriptive Measures; 4. Basic Probability	Location and dispersion, quantiles, association; events, conditional probability, independence and Bayes theorem.	Interactive lecture / practical R lab; Compute descriptive measures and basic probabilities in R.
4	5. Discrete Probability Distributions; 6. The Normal Distribution and Other Continuous Distributions	Binomial and Poisson models; normal, uniform and exponential distributions; probabilities and quantiles.	Interactive lecture / practical R lab; Evaluate, simulate and plot probability distributions in R.
5	7. Sampling Distributions; 8. Confidence Interval Estimation	Sampling variability, central limit theorem, standard errors; confidence intervals for means and proportions.	Interactive lecture / practical R lab; Simulate sampling distributions and estimate confidence intervals in R.
6	9. Fundamentals of Hypothesis Testing: One-Sample Tests	Hypotheses, p-values, significance, errors, assumptions, one-sample t and proportion tests.	Interactive lecture / practical R lab; Conduct and interpret one-sample tests in R.
7	10. Two-Sample Tests	Independent and paired comparisons; two-sample t tests and proportion tests.	Interactive lecture / practical R lab; Compare two groups using R and report effect size and

			uncertainty.
8	11. Analysis of Variance; 12. Chi-Square and Nonparametric Tests	One-way ANOVA and post-hoc analysis; chi-square goodness-of-fit and independence; rank-based tests.	Interactive lecture / practical R lab; Run ANOVA, chi-square and selected nonparametric tests in R.
9	13. Simple Linear Regression	Least squares; interpretation, residual diagnostics; confidence and prediction intervals.	Interactive lecture / practical R lab; Fit and diagnose a simple regression model in R.
10	14. Introduction to Multiple Regression	Partial effects, multiple predictors, dummy variables, model fit and coefficient interpretation.	Interactive lecture / practical R lab; Estimate and interpret a multiple regression model in R.
11	15. Multiple Regression Model Building	Interactions, multicollinearity, residual diagnostics, model selection and validation.	Interactive lecture / practical R lab; Build, compare and validate regression models in R.
12	16. Time-Series Forecasting	Trend, seasonality, moving averages, baseline forecasts and forecast error.	Interactive lecture / practical R lab; Generate and evaluate a short time-series forecast in R.
13	17. Business Analytics	Decision framing, descriptive and predictive analytics, communicating findings and responsible use of evidence.	Interactive lecture / practical R lab; Prepare an evidence-based business analytics case using R.
14	18. Getting Ready to Analyze Data in the Future	Reproducibility, ethical data analysis, analytical planning and emerging workflows.	Interactive lecture / practical R lab; Submit a reproducible analysis and reflect on limitations and responsible AI use.
15	Integration, review and applied synthesis	Integrate data preparation, exploratory analysis, statistical inference and modeling; final assessment preparation.	Interactive lecture / practical R lab; Present and review an integrated R analysis; practice interpreting statistical outputs.

The 15-week schedule begins with an introductory R module and includes all 18 statistical topics. Teaching weeks are indicative and subject to the approved academic calendar and contact hours.

Assessment Methods and Grading

Assessment	Proposed weight	Evidence
Assignments, R exercises and quizzes	30%	Individual applied problems and reproducible code
Midterm examination	30%	Conceptual and analytical problem solving
Final examination	40%	Integrated statistical analysis and interpretation

Total	100%	Subject to departmental approval

The course assessment distribution is 30% assignments and R exercises, 30% midterm examination, and 40% final examination. A capstone-style R report can be incorporated into coursework if approved.

Learning References and Resources

Suggested alternative statistics references (select any one or more, as available):

• Mendenhall, W., Beaver, R. J., and Beaver, B. M. Introduction to Probability and Statistics. Available for borrowing in the university library.

• Anderson, D. R., Sweeney, D. J., and Williams, T. A. Modern Business Statistics with Excel. Available in the university library.

• Berenson, M. L., Levine, D. M., and Krehbiel, T. C. Basic Business Statistics: Concepts and Applications.

R reference: Wickham, H., Çetinkaya-Rundel, M., and Grolemund, G. R for Data Science (2nd ed.). Free online: https://r4ds.hadley.nz/
• Grolemund, G. Hands-On Programming with R. Free online: https://rstudio-education.github.io/hopr/ (introductory R reference).

Software: R and RStudio (or equivalent R environment); recommended packages include tidyverse, ggplot2 and broom. Reading sections will be specified by topic; chapter numbering varies by textbook and edition.

Course Requirements and Policies

• Students should have computer access, internet connectivity, Moodle access and the ability to install or use R.

• Attendance, examination and academic-integrity policies follow current University of Petra regulations.

• Submitted work must identify sources, report methods transparently and distinguish individual work from permitted collaboration.

• AI use and disclosure: Students must declare any use of generative AI or other AI tools in assignments, projects, reports, code and other submitted coursework, whether or not AI use is expressly permitted. The declaration must name the tools used, describe how they were used, and identify affected sections or outputs. If no AI tools were used, students must state “No AI tools were used.” Undeclared or prohibited use is subject to University of Petra academic-integrity regulations; students remain responsible for the accuracy, originality and integrity of all submissions.

Assessment and Submission Guidance

R assignments should include the dataset/source, reproducible code, relevant tables/plots, interpretation and a statement of limitations. Submission dates, exam coverage and grading rubrics will be announced through Moodle.

Approval / Institutional Use

Role	Name / signature / date
Curriculum committee coordinator	[Pending]
Department head / Dean	[Pending]